BEING ETALE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A morphism is etale if and only if it is flat and G-unramified. Morphisms, Lemma morphisms-lemma-flat-unramified-etale, invoked at proof lines 6432-6433. Splits etaleness into two properties.
- Flatness and G-unramifiedness are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-flat and lemma-descending-property-unramified, invoked at proof lines 6434-6437. Gives both ingredients, hence the result.

Source reference. Original lemma 02L2: sources/stacks-descent/pinned-original-source.tex, source lines 6050–6050. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02VM: sources/stacks-descent/pinned-original-source.tex, source lines 6404–6404. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GV: sources/stacks-chow/tags.txt, source lines 3242–3242. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is étale" is fpqc local on the base.*

Proof. A morphism is étale if and only if it flat and G-unramified. See Morphisms, Lemma 02GV. We have seen already that being flat and G-unramified are fpqc local on the base (Lemmas 02L2, and 02VM). Hence the result follows. □

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